Exp No:9 Experiment to understand Matplotlib library use cases in Data Science through visualization

Description: Use Iris data set to understand the Matplot lib library

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_iris

iris = load_iris(as_frame=True)

df = iris.frame

#print(df.shape)

df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

 #  Column            Non-Null Count  Dtype

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 0  sepal length (cm)  150 non-null   float64

 1  sepal width (cm)   150 non-null   float64

 2  petal length (cm)  150 non-null   float64

 3  petal width (cm)   150 non-null   float64

 4  target           150 non-null   int64

dtypes: float64(4), int64(1)

memory usage: 6.0 KB
```

df.describe()

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

df.columns

Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',

'petal width (cm)', 'target'],

dtype='object')

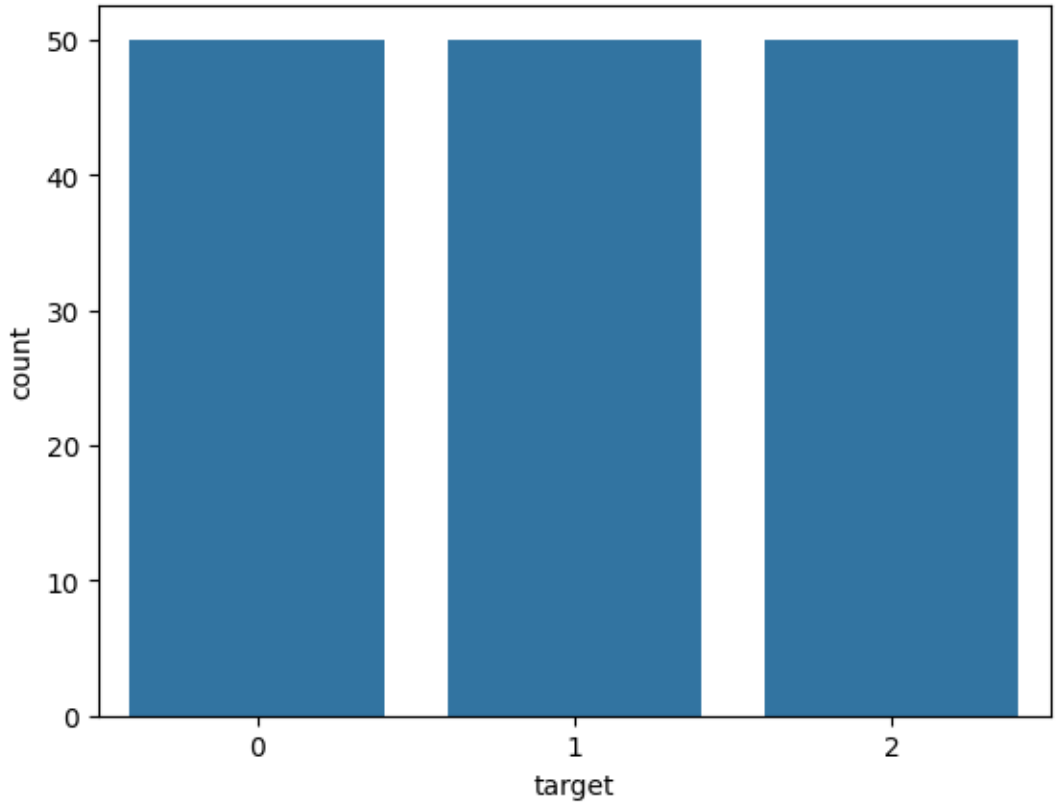
df.value_counts('target')

	count
target	
0	50
1	50
2	50

dtype: int64

sns.countplot(x='target',data=df)

plt.show()



dummies = pd.get_dummies(df.target)

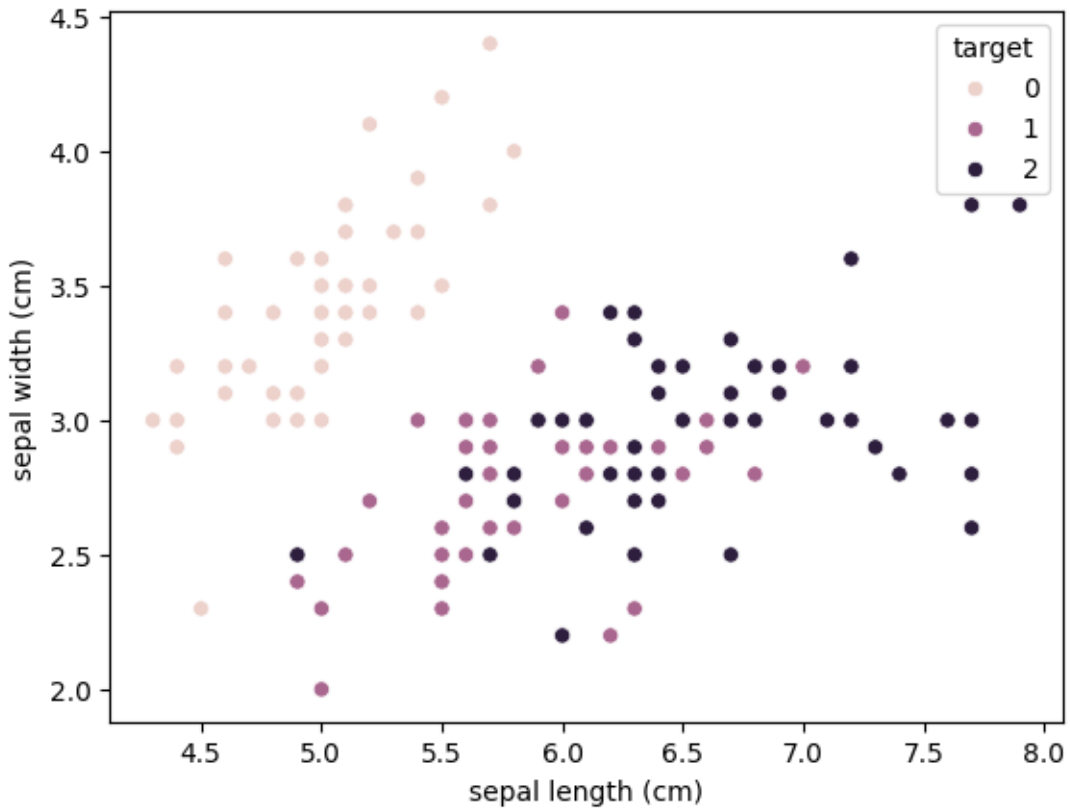
finaldataset = pd.concat([pd.get_dummies(df.target),df.iloc[:,[0,1,2,3]]],axis=1)

finaldataset.head()

	0	1	2	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	True	False	False	5.1	3.5	1.4	0.2
1	True	False	False	4.9	3.0	1.4	0.2
2	True	False	False	4.7	3.2	1.3	0.2
3	True	False	False	4.6	3.1	1.5	0.2
4	True	False	False	5.0	3.6	1.4	0.2

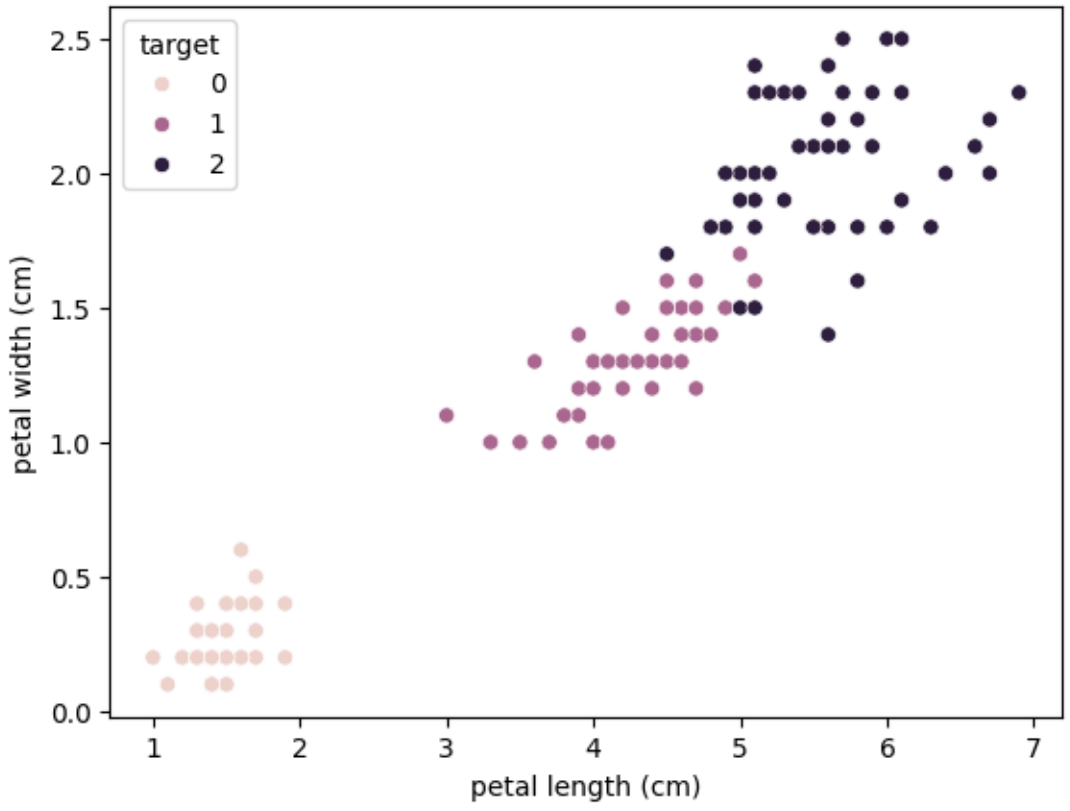
sns.scatterplot(x='sepal length (cm)',y='sepal width (cm)',hue = 'target',data=df)

<Axes: xlabel='sepal length (cm)', ylabel='sepal width (cm)'>



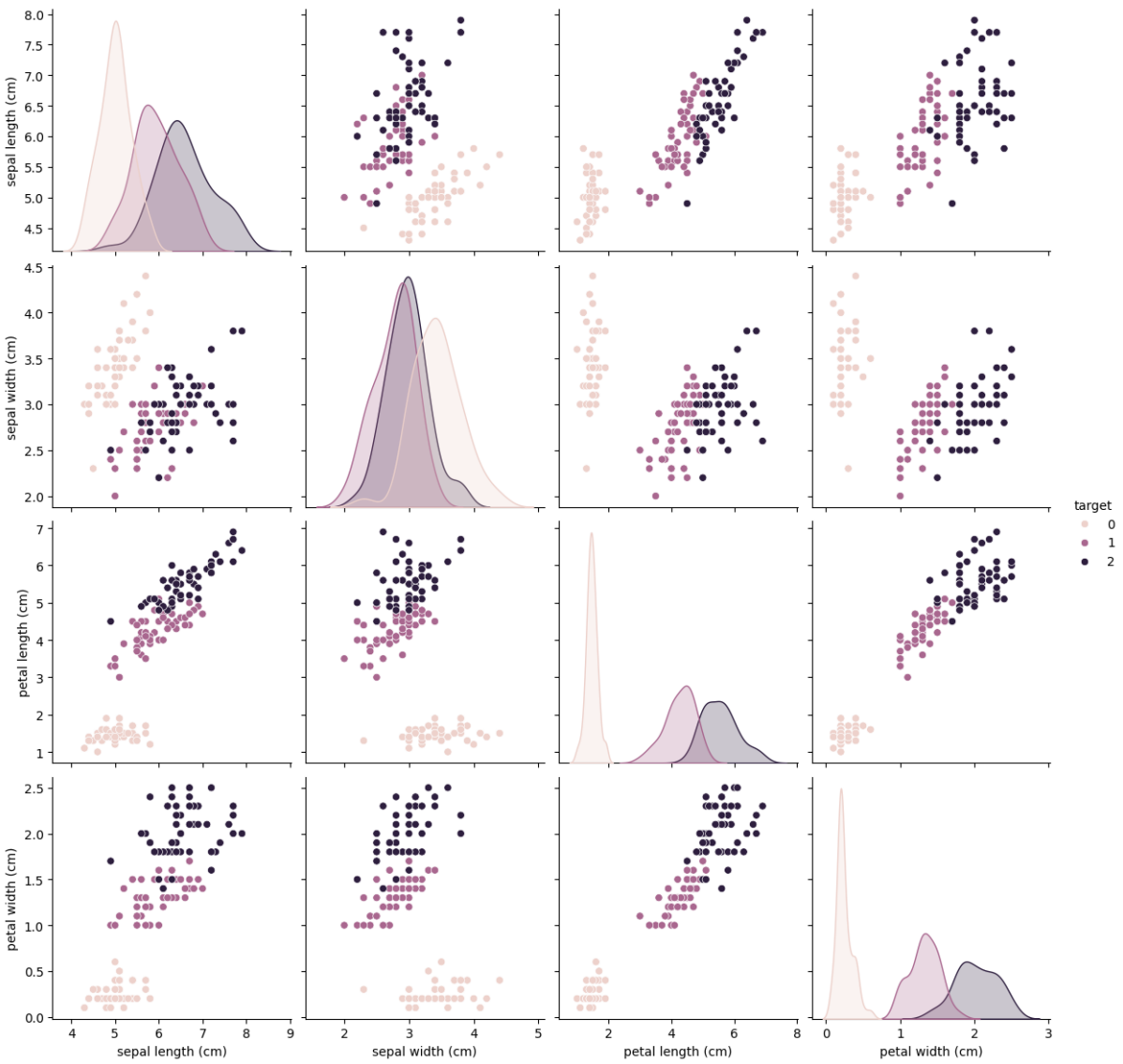
sns.scatterplot(x='petal length (cm)',y='petal width (cm)',hue = 'target',data=df)

<Axes: xlabel='petal length (cm)', ylabel='petal width (cm)'>



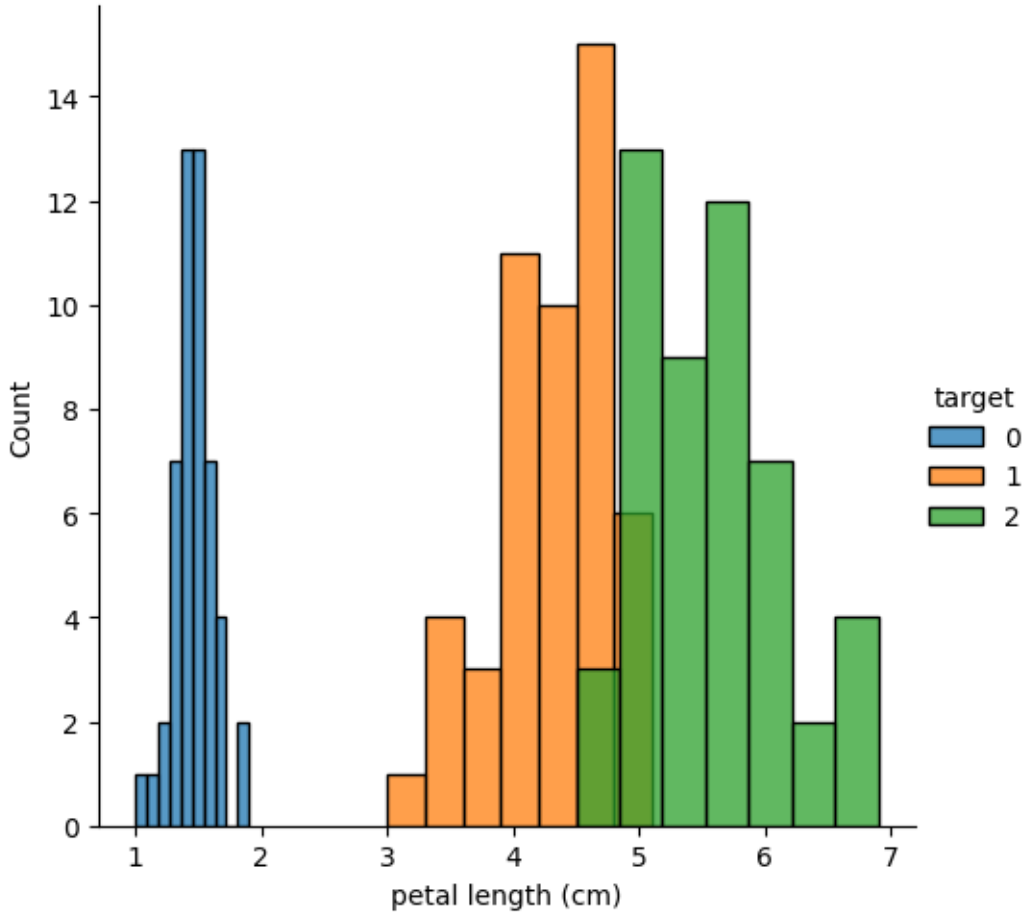
```
sns.pairplot(df,hue='target',height=3)
```

```
<seaborn.axisgrid.PairGrid at 0x7d92ffce2120>
```



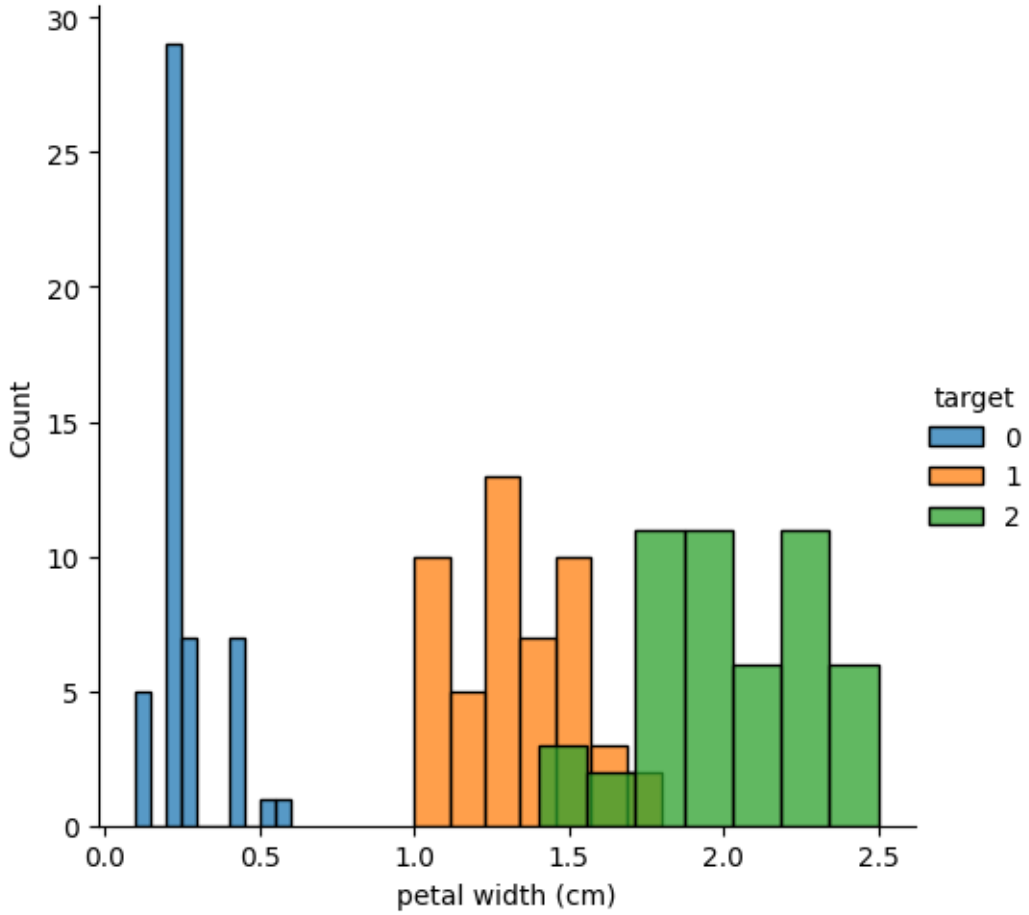

```
sns.FacetGrid(df,hue='target',height=5).map(sns.histplot,'petal length (cm)').add_legend()
```

```
plt.show()
```



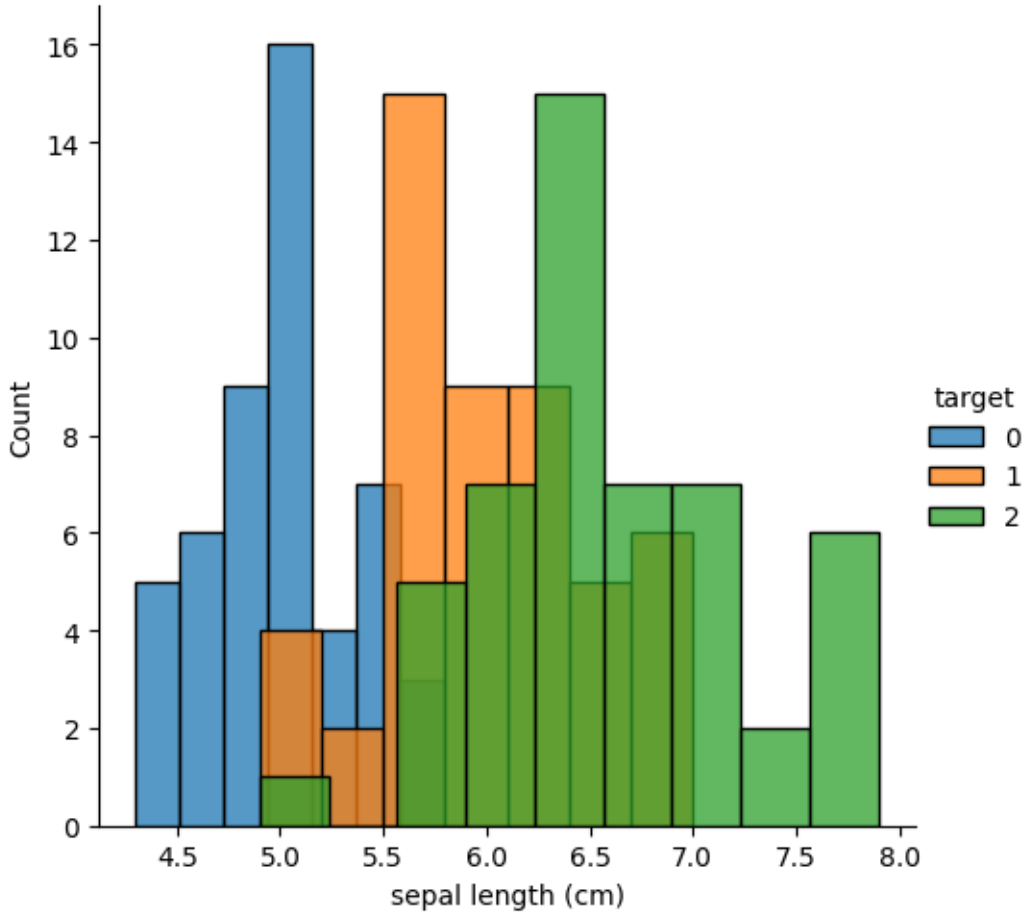

```
sns.FacetGrid(df,hue='target',height=5).map(sns.histplot,'petal width (cm)').add_legend()

plt.show()
```



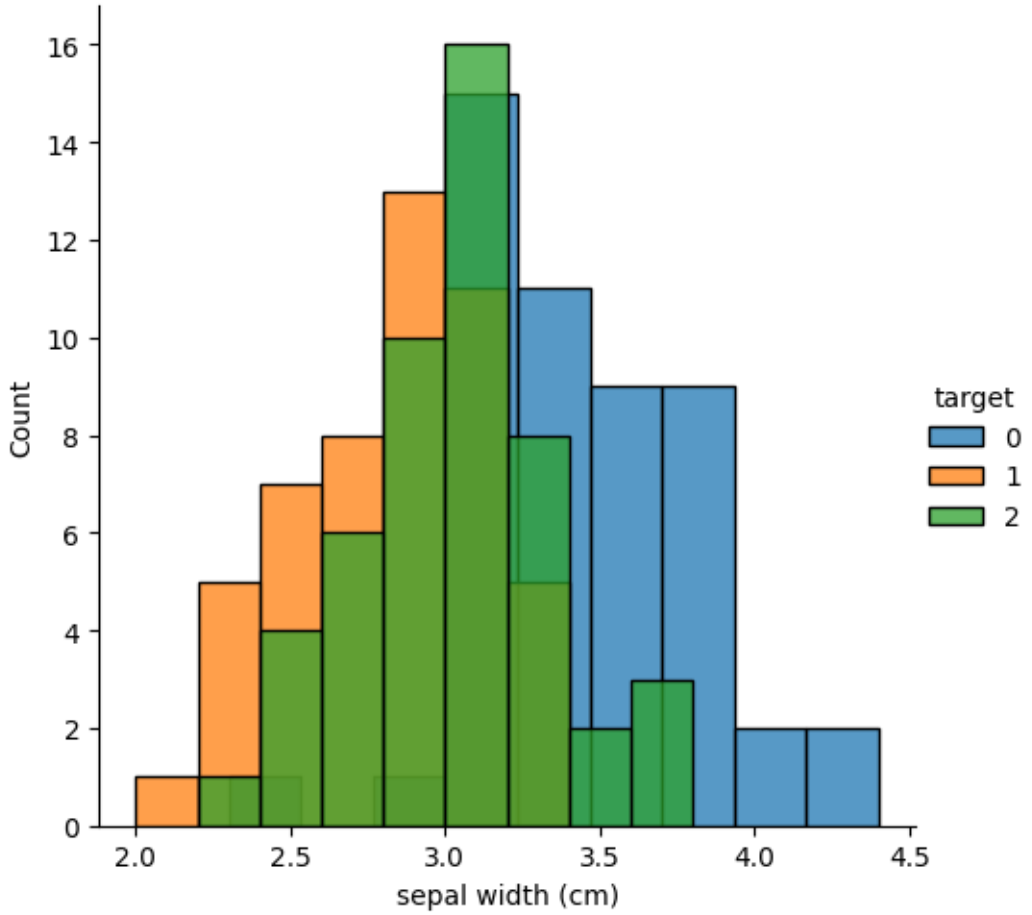

```
sns.FacetGrid(df,hue='target',height=5).map(sns.histplot,'sepal length (cm)').add_legend()

plt.show()
```




```
sns.FacetGrid(df,hue='target',height=5).map(sns.histplot,'sepal width (cm)').add_legend()

plt.show()
```



Result:

Iris data set has understand by using the Matplot library